

Typical Workflow Architecture Observed from Literature Survey

COMMON ACTORS

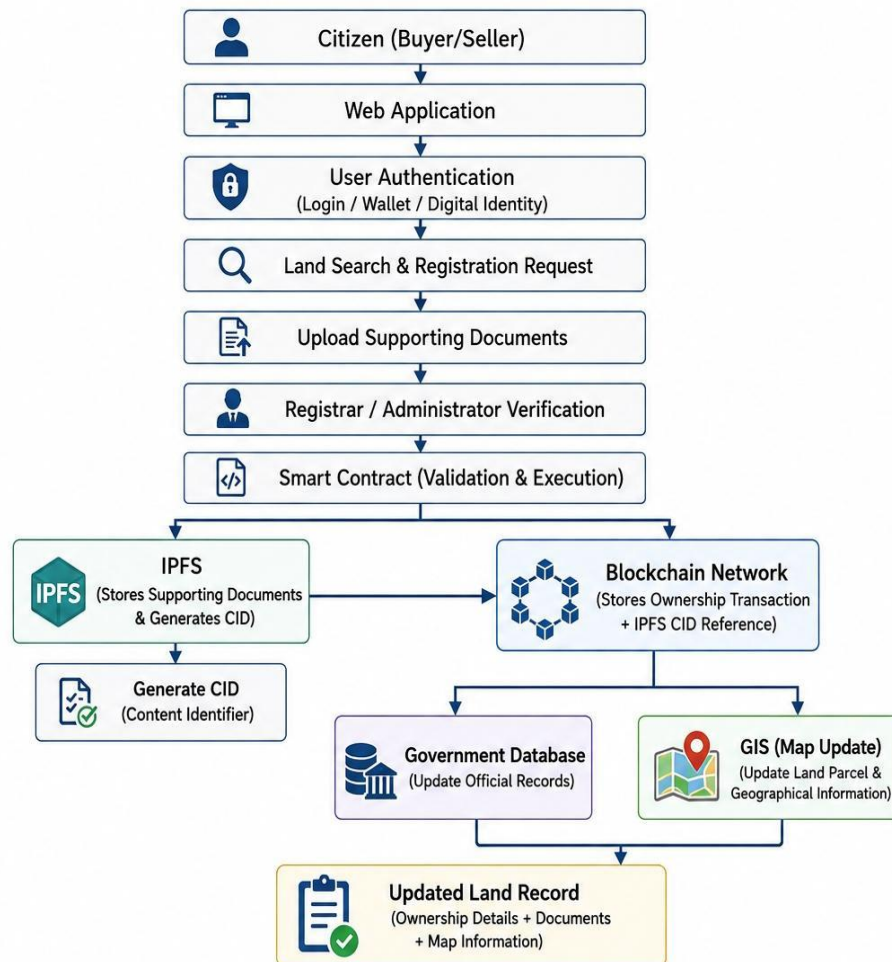
Actor	Typical Responsibility
Citizen (Buyer/Seller)	Searches land details, submits registration request, uploads required documents, requests ownership transfer.
Registrar / Administrator	Verifies documents, validates ownership, approves registration, updates official records.
Web Application	Common interface for citizens and government officials.
Smart Contract	Validates predefined rules and automates blockchain transactions.
Blockchain Network	Records approved ownership transactions and maintains immutable history.
IPFS (where implemented)	Stores supporting land documents securely using content identifiers (CID).
GIS (where implemented)	Displays survey maps, parcel boundaries, and geographical information.
Government Database (where implemented)	Maintains existing administrative and land records.

Our Observation from the Literature Survey

After reviewing the literature, the following observations were identified:

- Most implementation papers follow a similar workflow involving citizens, an administrative authority, smart contracts, blockchain, and supporting storage systems.
- Ethereum and Solidity are the most commonly adopted technologies for implementing blockchain-based land registration systems.
- IPFS is frequently integrated to store land documents, while blockchain maintains only the secure document reference (CID).
- Several studies include administrative verification before recording ownership on the blockchain, indicating that blockchain complements rather than replaces government authority.
- GIS is incorporated in some implementations to visualize land parcels and survey information.

Typical System Architecture Workflow Observed in Existing Blockchain-Based Land Record Management Systems



Important Note

Note: The government authority remains responsible for verifying and legally approving land ownership. Blockchain does not replace the government; instead, it securely records the approved transaction, while IPFS stores supporting documents and the government database and GIS maintain administrative and geographical records. Workflow Explanation

WORKFLOW EXPLANATION

Step 1: Citizen Access

The buyer or seller accesses the web application to search land information or initiate a land registration request.

Step 2: Web Application

The web application acts as the common interface through which citizens access land-related services such as land enquiry, registration requests, and document submission.

Step 3: User Authentication

For registration-related services, users authenticate using login credentials, a digital identity, or a blockchain wallet to verify their identity before performing secure transactions.

Step 4: Land Search & Registration Request

The citizen searches land details and submits a registration request by providing ownership information and other required details through the web application.

Step 5: Upload Supporting Documents

Supporting documents, such as sale agreements and ownership records, are uploaded to the system for verification and processing.

Step 6: Registrar / Administrator Verification

The registrar or authorized government administrator verifies the submitted documents, validates ownership details, and confirms that all legal requirements are satisfied before approving the registration.

Step 7: Smart Contract Validation & Execution

After administrative approval, the smart contract validates the predefined registration rules and executes the ownership transfer transaction on the blockchain network.

Step 8: IPFS Document Storage

The uploaded supporting documents are stored in IPFS, which generates a unique Content Identifier (CID). The actual documents remain in IPFS, while only the generated CID is referenced in the blockchain.

Step 9: Blockchain Transaction Recording

The blockchain records the approved ownership transaction together with the IPFS Content Identifier (CID), ensuring immutability, transparency, and tamper resistance.

Step 10: Government Database Synchronization

Once the blockchain transaction is successfully recorded, the government database is synchronized to update the official administrative land records.

Step 11: GIS Map Update

The GIS component updates the geographical information, including land parcel boundaries, survey details, and ownership information based on the approved transaction.

Step 12: Updated Land Record

The synchronized government database and GIS information together provide the updated land record, reflecting the latest ownership and land details.

Common Technologies Used

- Ethereum Blockchain
- Solidity Smart Contracts
- IPFS
- MetaMask
- React.js
- Node.js
- GIS
- Government Database Integration

Points for Further Evaluation Before Finalizing Our Architecture

These are discussion points, not project commitments:

1. Should public land enquiry require authentication, or only registration-related services?
2. Should wallet authentication be used only for ownership transactions?
3. What is the most suitable role of the registrar in a blockchain-based workflow?
4. At which stage should documents be stored in IPFS—before or after administrative approval?
5. How should blockchain synchronize with existing government databases?
6. Which blockchain architecture (Private or Consortium) is more appropriate for a government land registration system?